

The Galactic Canvas

An Interactive Storytelling Journey: Milky Way → Solar System → Sun → Planets

Research Compilation Document

Project Overview

The Galactic Canvas is an interactive storytelling website that takes the viewer on a visual and narrative journey — beginning with the Milky Way galaxy, zooming into our Solar System, moving toward the Sun, and finally exploring each planet in detail. The site is designed as a scroll- or zoom-based experience, guiding users from the largest cosmic scale down to individual planetary facts.

Section 1: Purpose, Audience & Rationale

Why Are We Using This Website?

This website is being created to make astronomy engaging and easy to understand through visual storytelling instead of plain text or static images. Space is a subject that is naturally vast and hard to grasp — by using an interactive zoom-based journey (Milky Way → Solar System → Sun → Planets), the site turns abstract scale and distance into something viewers can experience step by step. It combines education with an immersive, cinematic feel, helping users retain information better than a traditional textbook format, while also demonstrating skills in web design, interactivity, and data storytelling.

- To simplify a complex topic (the scale of the universe) through visual, interactive design
- To create an engaging, memorable learning experience rather than passive reading
- To demonstrate how storytelling techniques (zoom, scroll, narrative flow) can be used to teach science
- To showcase creative and technical skills — design, animation, and content research combined

Who Are We Using This Website For? (Target Audience)

- Students (school/college level) who want an engaging way to learn about space and astronomy
- Teachers/educators looking for a visual aid to explain scale, the Solar System, and planets in class
- Space enthusiasts and curious learners who enjoy interactive, self-paced exploration
- Evaluators/examiners of this assignment, who will assess both content accuracy and interactive design
- General website visitors with little prior astronomy knowledge — the content is written to be accessible, not overly technical

Why This Topic Was Chosen (Milky Way / Space)

- Space is a universally fascinating topic with strong visual potential, making it ideal for interactive storytelling
- It naturally supports a 'zoom-in' narrative structure — galaxy to star to planets — giving the project a clear beginning, middle, and end
- Reliable data (distances, sizes, compositions) is well-documented, making research and fact-checking straightforward
- The topic allows use of striking visuals, animations, and comparisons (scale, distance, temperature) that keep users engaged
- It has strong educational value, making the project useful beyond just an assignment — it can genuinely teach people something
- Personal interest in space and astronomy made the research and design process more motivating and enjoyable

Section 2: Research Questions (WH Questions)

Stage 1 — The Milky Way (Establishing Shot)

- What is the Milky Way, and what type of galaxy is it (structure-wise)?
- How big is the Milky Way (diameter, number of stars)?
- Where is our galaxy located in the universe, and what galaxies are nearby?
- Who discovered/named the Milky Way, and how did ancient cultures interpret it?
- How did the Milky Way form, and how old is it?
- What are the main parts of the galaxy (core, spiral arms, halo, disk)?
- Why does it look like a band of light across the night sky from Earth?
- What is at the center of the Milky Way (Sagittarius A*)?

Stage 2 — The Solar System (Zooming In)

- What is the Solar System, and where does it sit within the Milky Way?
- How did the Solar System form (nebular hypothesis)?
- How many objects make up the Solar System (planets, dwarf planets, moons, asteroids, comets)?
- Why do planets orbit the Sun the way they do?
- What holds the Solar System together?
- How long does it take the Solar System to orbit the galactic center?

Stage 3 — The Sun (The Zoom Target)

- What is the Sun made of, and what type of star is it?
- How does the Sun produce energy (nuclear fusion)?
- How old is the Sun, and what is its life cycle/future?
- How hot is the Sun (core vs. surface)?
- Why is the Sun important for life on Earth?
- What are solar phenomena (sunspots, solar flares, solar wind)?

Stage 4 — Each Planet (Template, repeat per planet)

- What is the planet made of (rocky / gas / ice giant)?
- How far is it from the Sun, and how long is its orbit (year length)?
- How long is a day on this planet (rotation period)?
- What is its size compared to Earth?
- Does it have moons — how many, and any notable ones?
- What is its atmosphere like, and surface temperature?
- Why is this planet unique (e.g., Saturn's rings, Mars's red color, Venus's greenhouse effect)?
- Who discovered it, or when was it first observed/visited by spacecraft?
- Has it been explored by any missions — what did we learn?

Bonus — Interactive / Design Research

- How have other space websites/apps handled zoom-scroll storytelling (for UX inspiration)?
- What tools/libraries support parallax or scroll-based zoom effects (Three.js, GSAP ScrollTrigger)?
- What accurate scale or distance comparisons could make the zoom feel real?

Section 3: Quantitative & Qualitative Data

Stage 1 — The Milky Way

Quantitative Data:

- Diameter: ~100,000–120,000 light-years
- Number of stars: 100–400 billion
- Age: ~13.6 billion years
- Solar System's distance from galactic center: ~26,000 light-years
- Thickness of the galactic disk: ~1,000 light-years
- Orbital speed around galactic center: ~828,000 km/h (1 orbit = ~230 million years — a 'galactic year')
- Number of known galaxies in the observable universe: ~2 trillion (estimated)

Qualitative Data:

- Classified as a barred spiral galaxy
- Made of a central bulge, spiral arms, a surrounding halo, and dark matter
- Contains gas, dust, and stellar nurseries where new stars form
- Named 'Milky Way' from its milky band appearance in the night sky (Latin: Via Lactea)
- Part of the 'Local Group' of galaxies, moving toward the Andromeda Galaxy (future collision predicted)

Stage 2 — The Solar System

Quantitative Data:

- Age: ~4.6 billion years
- Number of planets: 8 (plus 5 recognized dwarf planets)
- Diameter of the Solar System (to Oort Cloud): ~2 light-years
- Number of known moons: 300+ across all planets
- Asteroid belt: over 1 million asteroids larger than 1 km

Qualitative Data:

- Formed from a collapsing giant molecular cloud (nebular hypothesis)
- Structure: inner rocky planets, asteroid belt, outer gas/ice giants, Kuiper Belt, Oort Cloud
- Held together by the Sun's gravity
- Planets orbit in roughly the same plane (ecliptic plane) and direction

Stage 3 — The Sun

Quantitative Data:

- Diameter: ~1.39 million km (109x Earth's diameter)
- Mass: 99.86% of the Solar System's total mass
- Surface temperature: ~5,500°C
- Core temperature: ~15 million°C
- Age: ~4.6 billion years; expected lifespan ~10 billion years
- Composition: ~73% hydrogen, ~25% helium

Qualitative Data:

- Classified as a G-type main-sequence star ('yellow dwarf')
- Powered by nuclear fusion (hydrogen fusing into helium)

- Exhibits phenomena like sunspots, solar flares, and solar wind
- Will eventually become a red giant, then a white dwarf

Stage 4 — Planets: Quantitative Reference Table

Planet	Distance from Sun (M km)	Diameter (km)	Day length	Year length	Moons
Mercury	57.9	4,879	59 days	88 days	0
Venus	108.2	12,104	243 days	225 days	0
Earth	149.6	12,742	24 hrs	365.25 days	1
Mars	227.9	6,779	24.6 hrs	687 days	2
Jupiter	778.5	139,820	9.9 hrs	11.9 years	95
Saturn	1,434	116,460	10.7 hrs	29.5 years	146
Uranus	2,871	50,724	17.2 hrs	84 years	27
Neptune	4,495	49,244	16.1 hrs	165 years	14

Qualitative Highlights (for storytelling flavor):

- Mercury — smallest planet, extreme temperature swings, cratered like the Moon
- Venus — hottest planet due to runaway greenhouse effect, rotates backward
- Earth — only known planet with life, liquid water, protective atmosphere
- Mars — 'Red Planet' due to iron oxide dust, evidence of ancient water, target for human exploration
- Jupiter — largest planet, Great Red Spot storm, strong magnetic field
- Saturn — iconic ring system made of ice and rock, least dense planet (would float in water)
- Uranus — rotates on its side (~98° tilt), pale blue-green from methane
- Neptune — strongest winds in the Solar System, deep blue color, discovered through mathematical prediction

Note: Moon counts (especially for Jupiter and Saturn) update occasionally as new moons are confirmed — worth a quick check closer to submission for precision.

Section 4: Project Name

Chosen Name:

The Galactic Canvas

This name reflects the site's core concept — presenting the vastness of the galaxy, the Solar System, and the planets as an unfolding visual and narrative canvas for the viewer to explore.